Exploratory Data Analysis (EDA) Summary Report

1. Introduction

The purpose of this report is to analyze customer demographic, financial, and credit-related data in order to evaluate creditworthiness, repayment behavior, and default risk. By consolidating key indicators such as income, loan balance, credit score, repayment ratio, and missed payments, the report provides a comprehensive view of the loan portfolio and customer risk profiles.

Goals of the Report

1. Credit Risk Assessment

- Identify customers with a high likelihood of default.
- Evaluate the relationship between income, loan balance, credit score, and repayment behavior.
- Strengthen credit approval processes through data-driven insights.

2. Portfolio Monitoring

- Track the overall health of the loan book, including repayment ratios and default trends.
- Detect early warning signals from customer behavior, such as declining repayment ratios or repeated missed payments.

3. Customer Segmentation

- Classify customers into risk categories based on demographic and financial indicators.
- Support targeted lending strategies by distinguishing between low-risk and high-risk segments.

4. Operational Efficiency

- Improve collection strategies by focusing on customers with repayment challenges.

- Reduce non-performing assets (NPAs) and associated financial risks.
- Ensure alignment with regulatory and compliance standards.

5. Strategic Decision-Making

- Optimize loan offerings, interest rates, and repayment terms according to risk categories.
- Enable financial inclusion by identifying low-risk yet underserved customer groups.
- Support business growth through informed marketing and customer engagement strategies.

The dataset is to monitor and analyze the credit risk and payment behavior.

2. Dataset Overview

The total number of rows is approximately 200 and total number of columns is 17.

Income, Loan balance, Credit score and missed payments are missing in some rows.

Anomalies that I found:

Credit score needs validation.

Repayment ratio must between 0 and 1.

Employment status must include possible inconsistent categories.

Gender should only contain male and female.

Missed payments should be an integer greater than 0.

Monthly payment columns should be limited to paid, missed or late.

3. Missing Data Analysis

Customer_ID	Unique customer identifier	String/Integer
Age	Customer age	Integer
Gender	Male/Female	Categorical
Income	Annual income	Numeric (Float/Int)
Education	Education level	Categorical
Employment_Status	Employment status	Categorical
Loan_Amount	Approved loan amount	Numeric (Float/Int)
Loan_Balance	Outstanding balance	Numeric (Float/Int)
Credit_Score	Creditworthiness score	Integer

	(range 300–850 typical)	
Default_Flag	Whether customer defaulted (0 = No, 1 = Yes)	Binary Integer
Repayment_Ratio	Repayment made vs. expected (0–1)	Float
Missed_Payments_3M	Count of missed payments in last 3 months	Integer
Month_1 … Month_6	Monthly repayment status (Paid / Missed / Late)	Categorical

Income-Some rows have blank values (e.g., CUST0041, CUST0043, CUST0060, CUST0144)

Loan_Balance-Some rows have blanks (e.g., CUST0009, CUST0024, CUST0026, CUST0050, CUST0053, CUST0103)

Credit_Score-Rarely missing (e.g., CUST0116)

Customer_ID-One row seems empty (row 129) → possible misalignment

Monthly repayment columns (Month_1 … Month_6)-Usually complete

Other columns (like Age, Employment_Status, Education) appear mostly complete.

4. Key Findings and Risk Indicators

Credit_Score vs Missed_Payments	Negative correlation (lower score → more missed payments)
Debt_to_Income vs Delinquent_Account	Positive correlation (higher debt burden → higher risk)
Income vs Missed_Payments	Slight negative correlation (higher income → fewer missed payments)
Employment_Status vs Delinquency	Unemployed/self-employed → higher missed payments
Age vs Missed_Payments	Young age → higher risk, older → mixed trend
Credit_Utilization vs Missed_Payments	Positive correlation, but some anomalies in low-utilization group

5. AI & GenAI Usage

Prompts Used:

“summarize and analyze the given data.Also clean the duplicate data and replace the missing values with most appropriate method”

“Analyze the dataset and find the purpose and goals of this dataset.”

“Analyze and find total number of rows and columns. Also find the appropriate anomalies.”

“Find the key risk indicators of this dataset.’

6. Conclusion & Next Steps

The dataset analysis indicates that customer delinquency is strongly associated with low credit scores, high debt-to-income ratios, elevated credit utilization, unemployment or self-employment, and repeated missed payments. Younger customers and those with lower incomes are also at higher risk, while anomalies such as missing Loan_Balance, blank Customer_IDs, and outlier values highlight data quality issues that could affect insights. Overall, repayment patterns over six months provide clear signals for identifying high-risk customers.